

Faculty of Computer Technology and Cybersecurity
Department of Computer Engineering
ARCHITECTURE AND ORGANIZATION OF COMPUTER SYSTEMS
LABORATORY WORK - 2

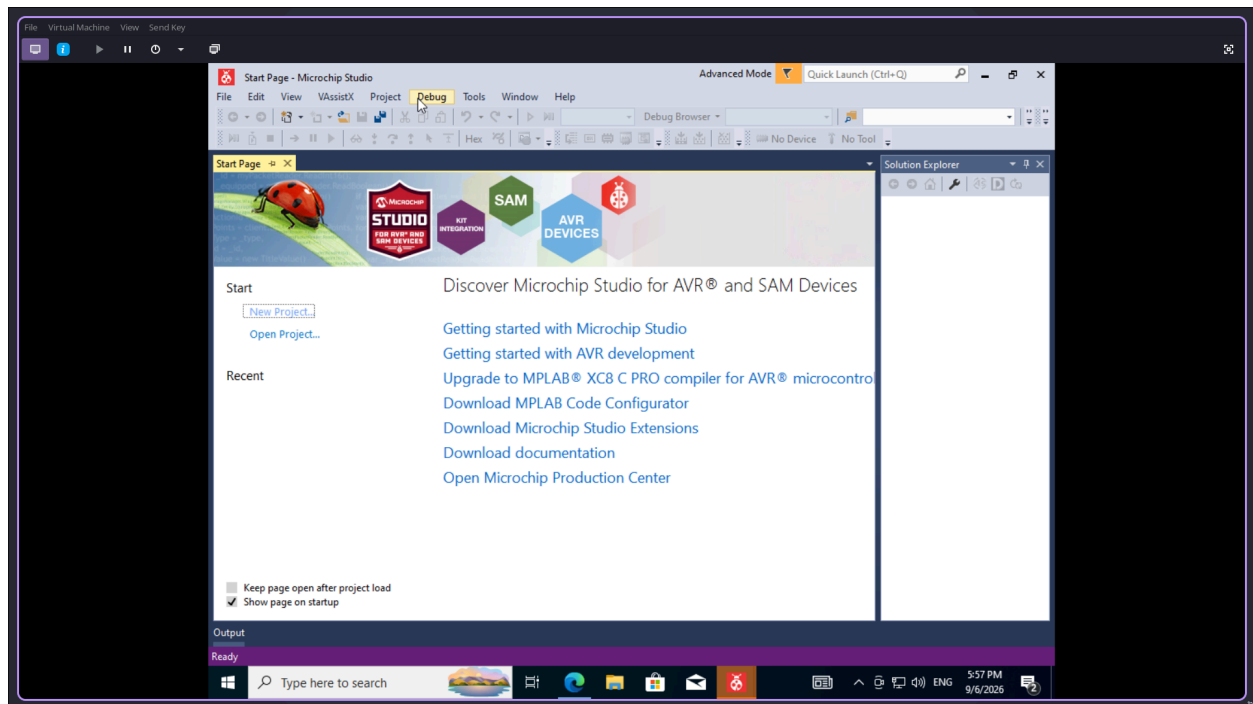
Date: 19.09.2026

Full Name: Daniil Kalts

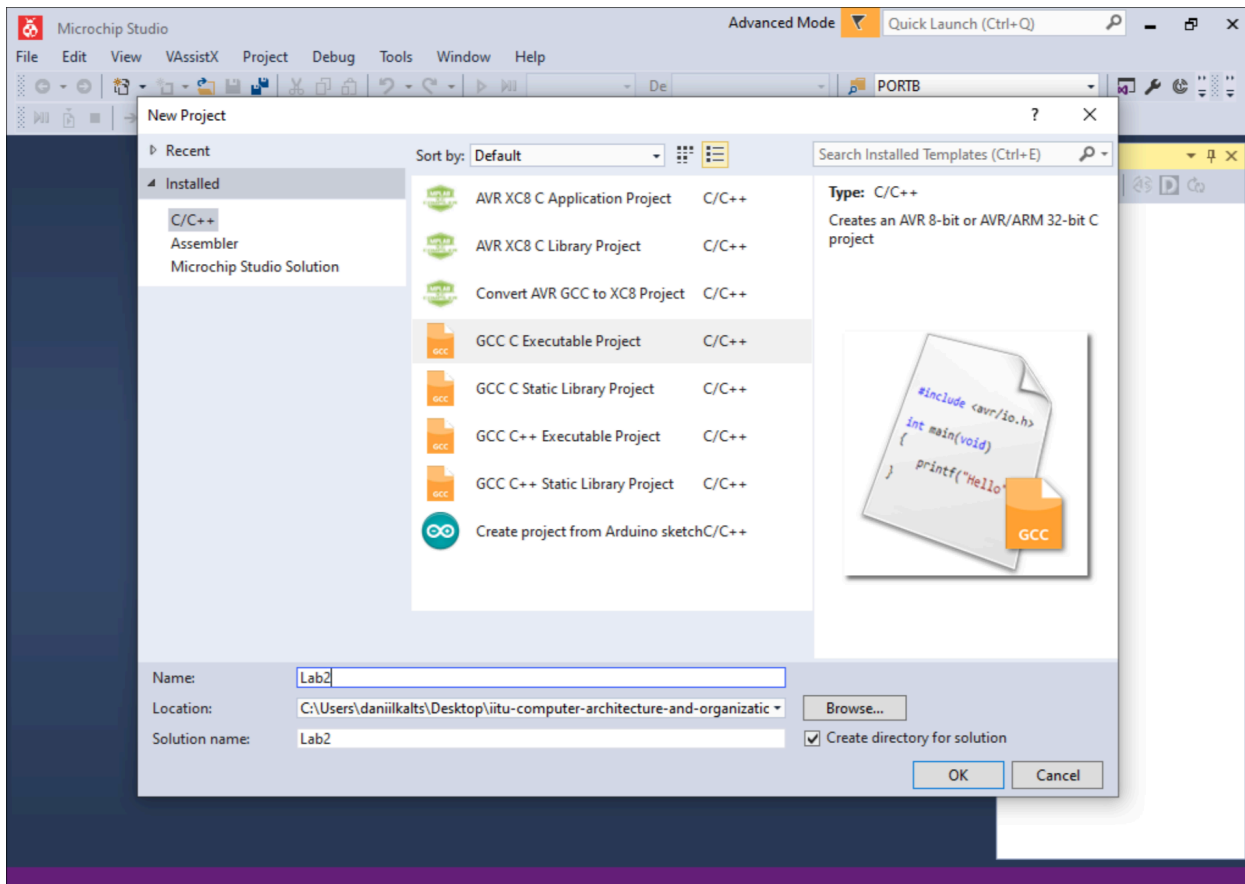
Group: IT2-2404SE

PROGRESS OF WORK

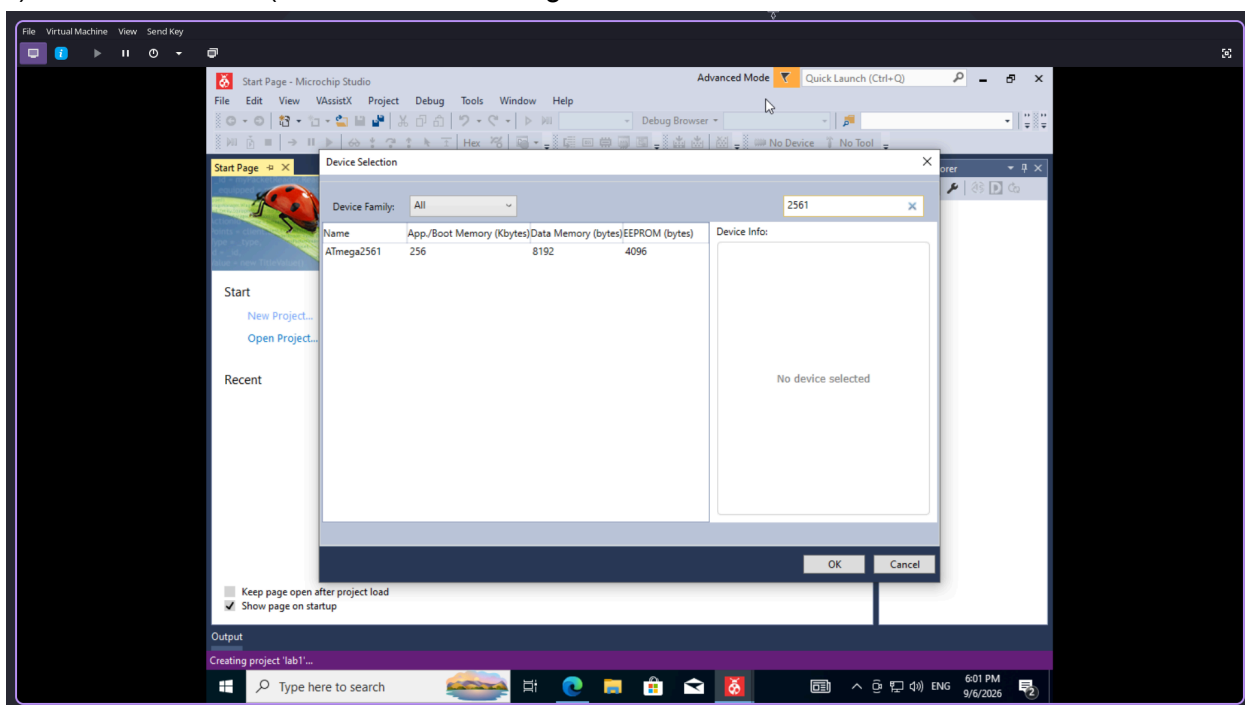
1) Start the Microchip Studio.



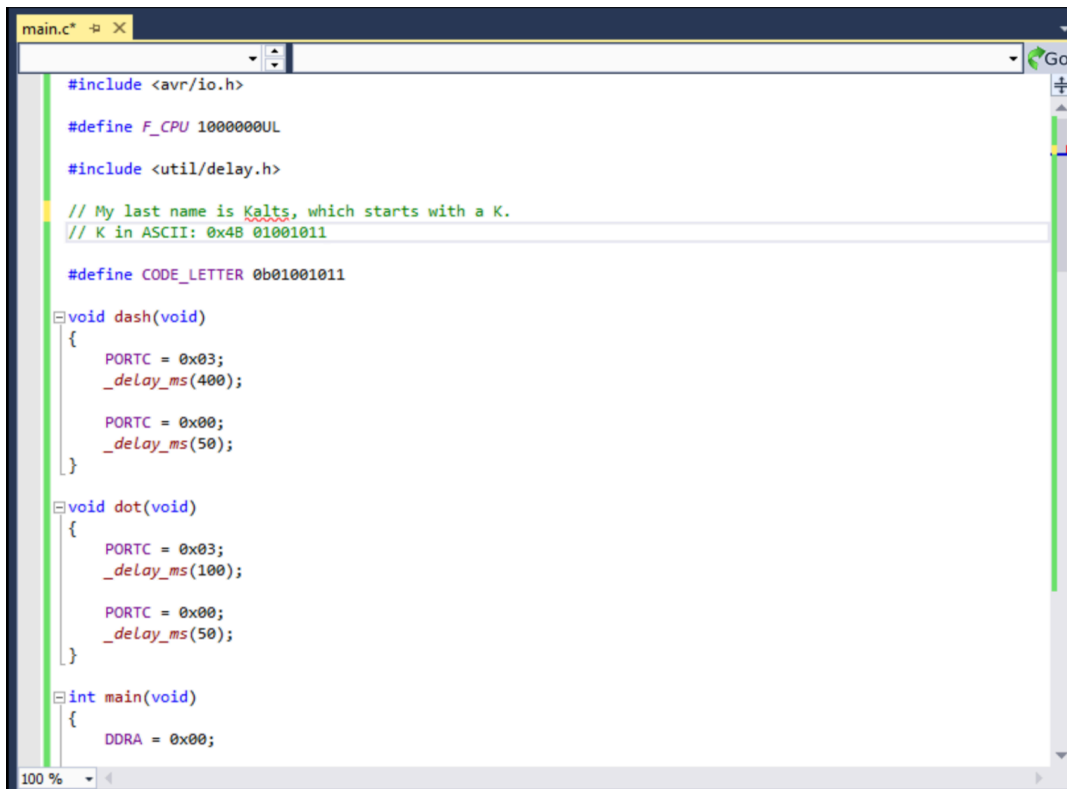
2) Create a new GCC C Executable project.



3) Select the device (in this work — ATmega2561).



4) Next, program the microcontroller.



```
main.c* X
#include <avr/io.h>

#define F_CPU 1000000UL

#include <util/delay.h>

// My last name is Kalts, which starts with a K.
// K in ASCII: 0x4B 01001011

#define CODE_LETTER 0b01001011

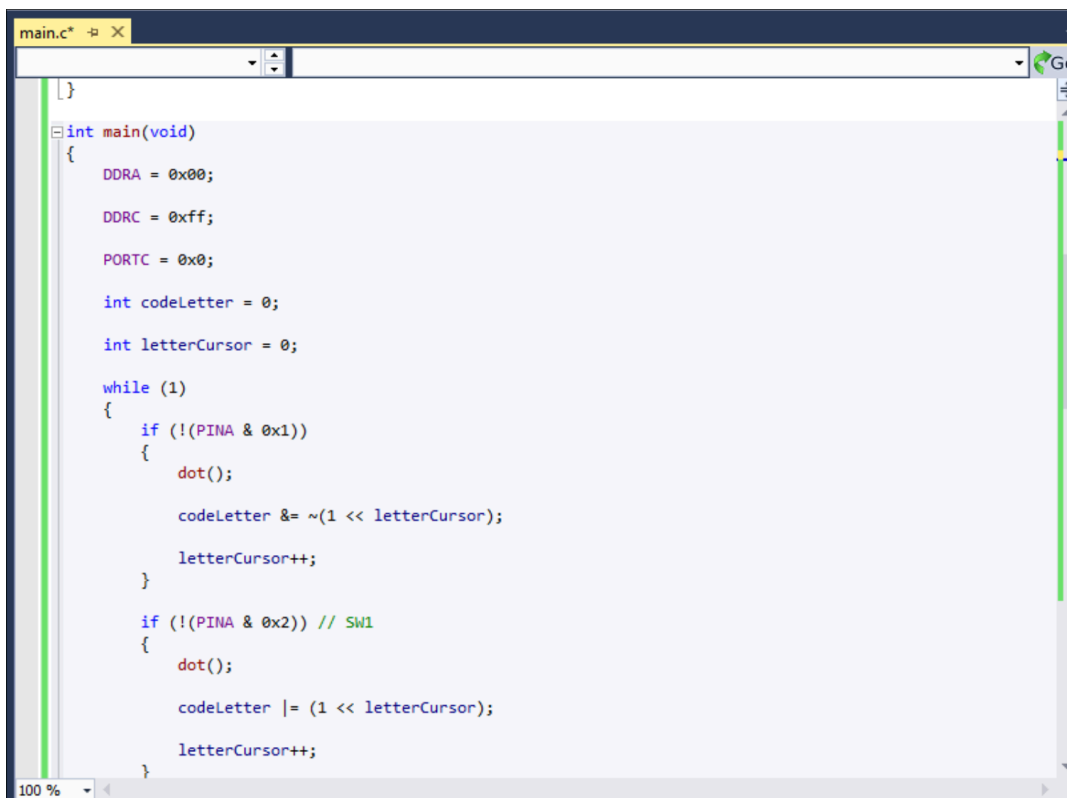
void dash(void)
{
    PORTC = 0x03;
    _delay_ms(400);

    PORTC = 0x00;
    _delay_ms(50);
}

void dot(void)
{
    PORTC = 0x03;
    _delay_ms(100);

    PORTC = 0x00;
    _delay_ms(50);
}

int main(void)
{
    DDRA = 0x00;
```



```
main.c* X
}

int main(void)
{
    DDRA = 0x00;

    DDRC = 0xff;

    PORTC = 0x0;

    int codeLetter = 0;

    int letterCursor = 0;

    while (1)
    {
        if (!(PINA & 0x1))
        {
            dot();

            codeLetter &= ~(1 << letterCursor);

            letterCursor++;
        }

        if (!(PINA & 0x2)) // SW1
        {
            dot();

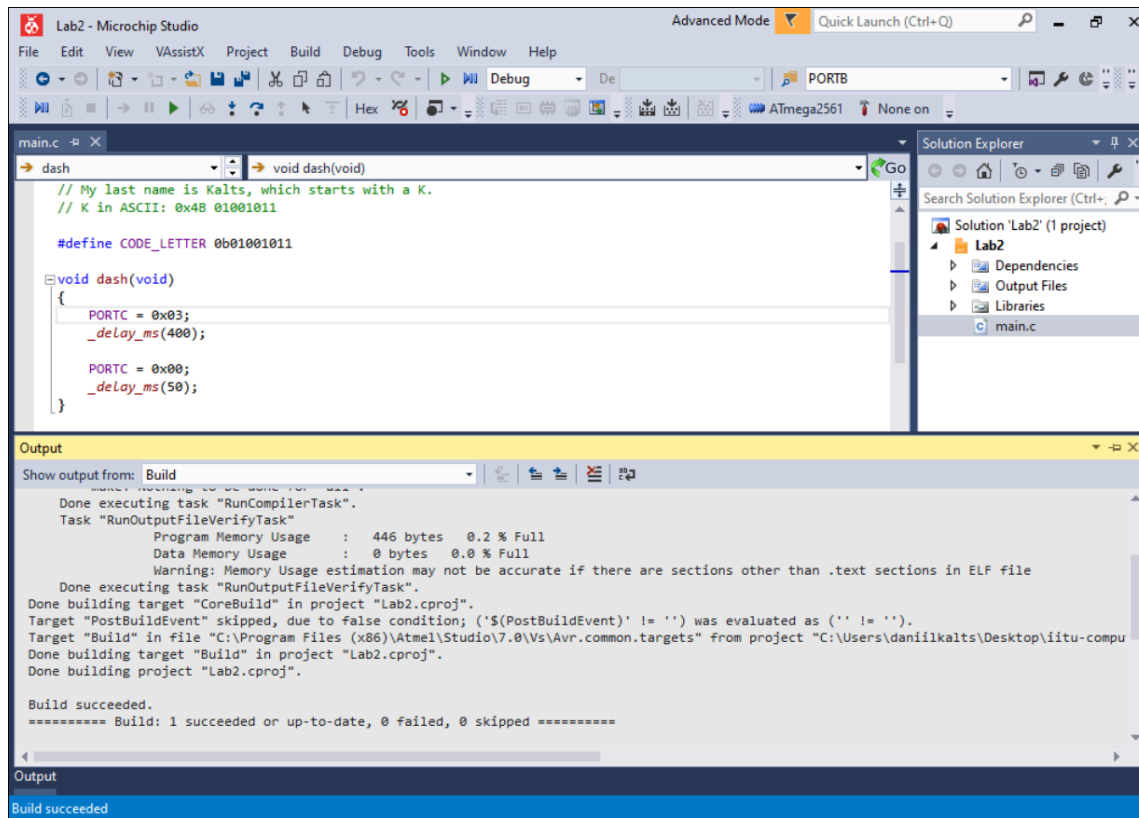
            codeLetter |= (1 << letterCursor);

            letterCursor++;
        }
    }
}
```

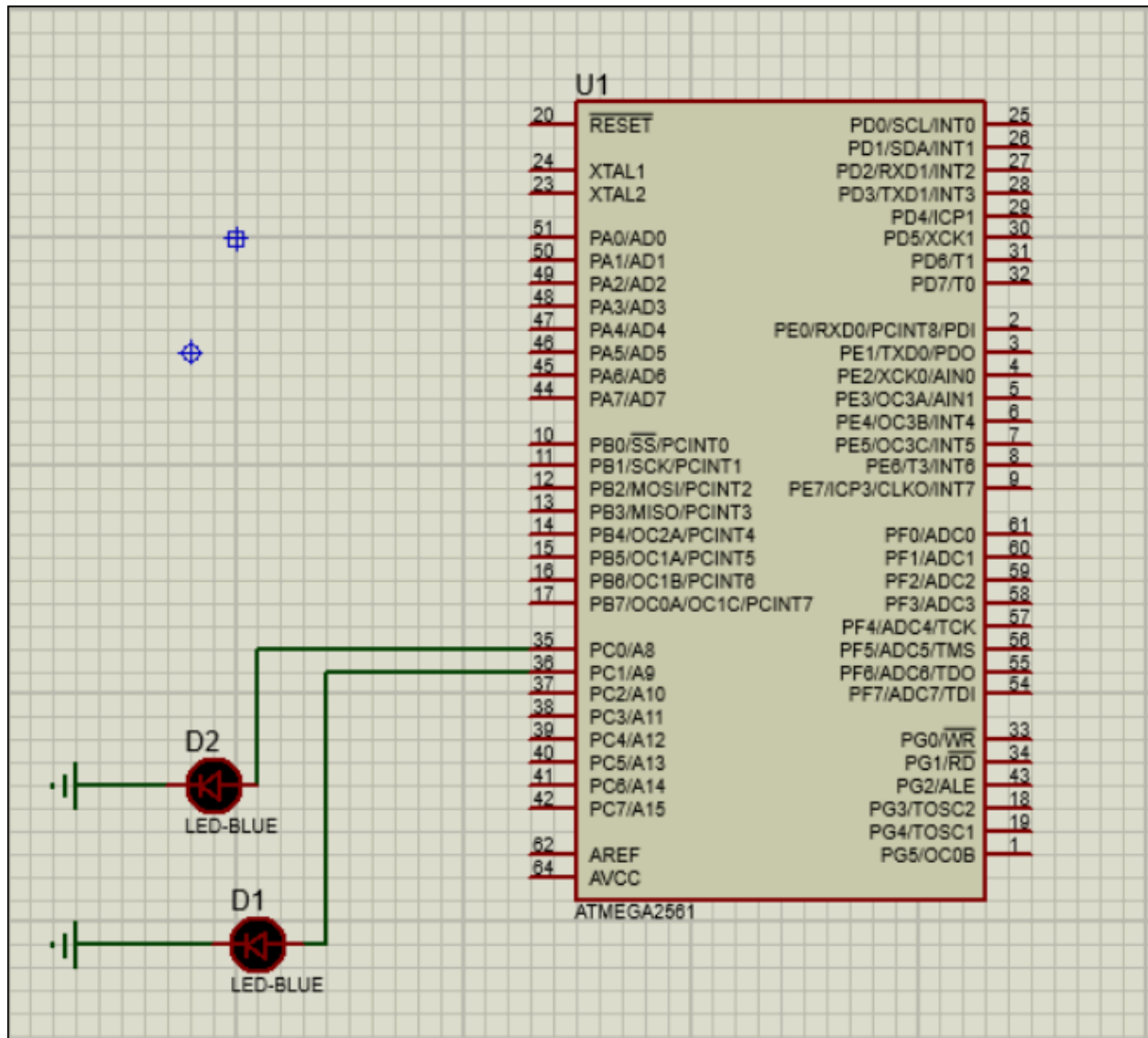
```
main.c ✕  
if (!(PINA & 0x4)) // SW2  
{  
    dash();  
  
    letterCursor = 0;  
  
    codeLetter = 0;  
}  
  
if (letterCursor >= 8)  
{  
    if (codeLetter == CODE_LETTER)  
    {  
        dot();  
  
        dot();  
  
        dot();  
    }  
    else  
    {  
        dash();  
  
        dash();  
  
        dash();  
    }  
  
    letterCursor = 0;  
}
```

```
main.c ✕  
if (letterCursor >= 8)  
{  
    if (codeLetter == CODE_LETTER)  
    {  
        dot();  
  
        dot();  
  
        dot();  
    }  
    else  
    {  
        dash();  
  
        dash();  
  
        dash();  
    }  
  
    letterCursor = 0;  
  
    codeLetter = 0;  
}  
  
    _delay_ms(100);  
}  
  
return 0;  
}
```

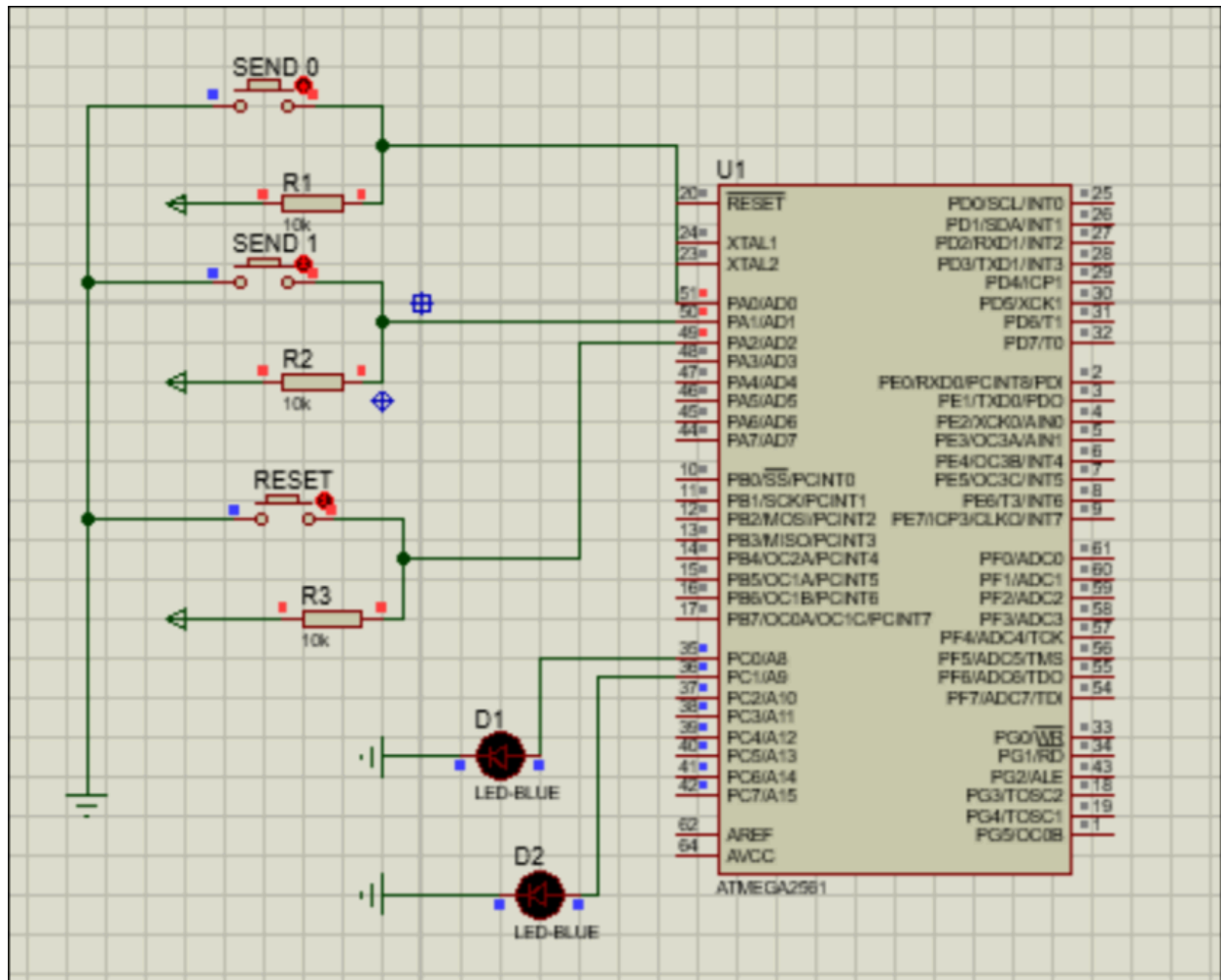
5. Compile the project and obtain the .hex file.



6. Open Proteus (ISIS Professional) and pick the ATmega2561, LED components and ground them.



7. Add the remaining components: three buttons and three resistors.

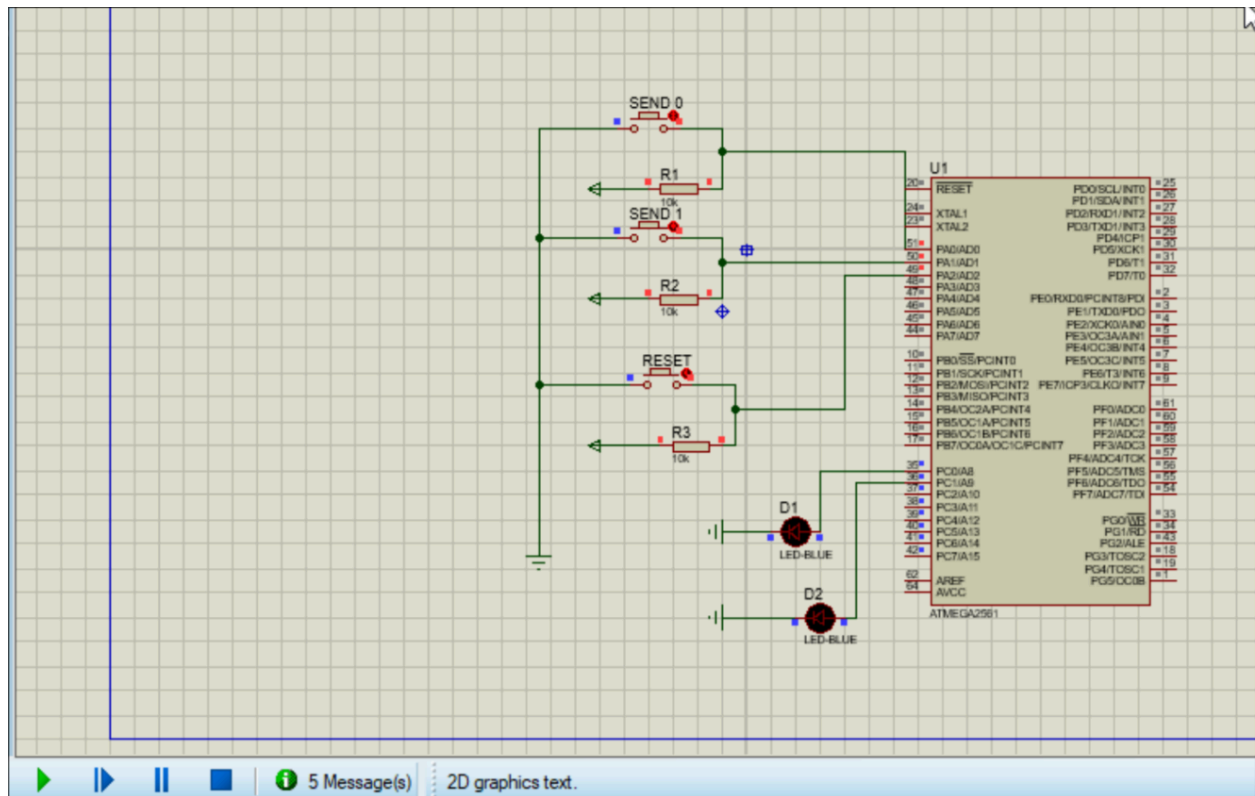


8. Load the compiled .hex file into the microcontroller's properties (Program File) and configure the component

Edit Component [?] [X]

Part <u>R</u> eference:	U1	Hidden: ☐
Part <u>V</u> alue:	ATMEGA2561	Hidden: ☐
<u>E</u> lement:	New	
PCB Package:	QFP80P1600X1600X120-6 AA	Hide All ▾
Program File:	..\1-week\Lab2\Lab2\Debug\ Folder	Hide All ▾
CLKDIV8 (Divide clock by 8)	(0) Programmed ▾	Hide All ▾
CKOUT (Clock output)	(1) Unprogrammed ▾	Hide All ▾
WDTON (Watchdog timer always on)	(1) Unprogrammed ▾	Hide All ▾
BOOTSRT (Select Reset Vector)	(1) Unprogrammed ▾	Hide All ▾
CKSEL Fuses:	(0010) Int. RC Osc. 8MHz ▾	Hide All ▾
Boot Loader Size:	(00) 4096 words. Starts at 0x1F0 ▾	Hide All ▾
SUT Fuses:	(00) ▾	Hide All ▾
Advanced Properties:		
Clock Frequency ▾	(Default) ▾	Hide All ▾
Other <u>P</u> roperties:	^ ▾	
☐ Exclude from Simulation ☐ Attach hierarchy module ☐ Exclude from PCB Layout ☐ Hide common pins ☐ Exclude from Current Variant ☐ Edit all properties as text		

9. Run the simulation and verify that the entered ASCII code is checked correctly using the LED signals.



Conclusion:

In this laboratory work, I programmed the ATmega2561 microcontroller in C and tested the circuit in Proteus. Port A was configured to receive the binary ASCII code of the letter K through three buttons, while Port C controlled two LEDs. The program confirmed each entered bit with a short signal, reset the input with a long signal, and indicated whether the complete code was correct or incorrect.